

Hamza Ouriour

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EXPERIENCE

NASA May – Sep 2025
Robotics Engineering Intern Houston, TX

- Assembled and integrated hardware for the Lunar Terrain Vehicle Ground Test Unit (LTV GTU), a full-scale testbed supporting NASA's Artemis Mission crew accommodation and lunar surface technology evaluation.
- Managed part procurement, revision cycles, and component-level assembly across 6+ LTV subsystems; maintained configuration traceability through iterative build cycles.
- Coordinated testbed setup, functional verification, and ongoing maintenance with a cross-functional team to ensure system readiness for lunar simulation testing.

NASA (Europa Clipper Mission) Jun 2024 – Jan 2025
EIS Optimization Engineer Intern Hybrid

- Developed autonomous imaging optimization algorithms for Europa Clipper's Europa Imaging System (EIS), enabling intelligent science-target prioritization during planetary flyby windows.
- Improved frame-selection efficiency by ~23% via refined decision heuristics; contributed to pre-launch validation and performance testing of flight-critical imaging software.
- Supported cross-functional validation efforts with mission scientists and software engineers to verify algorithm behavior under simulated flyby scenarios.

AeroSense PHM Oct 2025 – Present
Founder & Lead Engineer Boston, MA

- Engineered end-to-end ML pipeline (Random Forest + Gradient Boosting) across 17 physics-grounded features; achieved 94.2% fault classification accuracy, 97% critical-fault recall, and 64.2-cycle RUL RMSE across 5 benchmarked models.
- Founded a full-stack Prognostics & Health Management (PHM) SaaS platform for BLDC/ball-screw electro-mechanical actuators on commercial aircraft, targeting a gap in the \$26,000-aircraft global fleet.
- Built 7-layer system architecture spanning physics simulation (BLDC dynamics, Weibull/Arrhenius degradation), feature extraction, Flask REST API, SQLite persistence, and MRO dashboard; validated with 39/39 passing tests.

Hallam-ICS Dec 2025 – Present
Mechanical Engineer Intern Boston, MA

- Designed fabrication-ready layouts and construction drawings for semiconductor and research facilities using Revit and AutoCAD.
- Developed mechanical/utility system layouts (process gas, cooling water, exhaust) and supported P&ID and equipment cut-sheet reviews.
- Coordinated with senior engineers on code-compliant system design, contributing to facility commissioning documentation and design review cycles.

Starship Robotics Aug 2023 – Apr 2024
Robotics Engineer & Data Scientist Boston, MA

- Developed Python/C++ control software for autonomous navigation; integrated CV/ML pipelines for real-time object detection and motion planning across 4 robotic platforms.
- Designed and validated embedded electronics for sensor-actuator systems, optimizing closed-loop performance and reliability through iterative hardware testing.
- Analyzed operational sensor data to identify failure patterns and improve system reliability; produced performance reports to guide hardware and software iterations.

PROJECTS

Autonomous Precision UAV Landing & Navigation System

PX4 SITL | Gazebo | ROS | OpenCV | MAVSDK | Python | Ubuntu

- Engineered a vision-guided autonomous landing pipeline for GPS-denied quadrotor operations in PX4 SITL/Gazebo; achieved <10 cm landing dispersion across 70+ simulation runs.
- Redesigned obstacle avoidance and path planning logic, increasing standoff distance from obstacles by ~40% and eliminating trajectory oscillation through velocity setpoint smoothing.
- Enhanced LiDAR/vision perception stack for more reliable autonomous navigation; resolved NED/ENU frame inconsistencies for stable 2 m hover under simulated crosswind disturbances.

Thermofluids Cycle Visualizer

Python | NumPy | Matplotlib

- Modeled Rankine and Brayton thermodynamic cycles with <2% deviation from reference efficiencies; automated P–V/T–S diagram generation and parametric pressure/temperature sweeps.

Laser Head-Tracking Robot

Arduino | C++ | OpenCV | Servo Control

- Built a 180° real-time vision-tracking system with PID angular correction; cut tracking latency by 35% via loop timing optimization; deployed with custom PCB wiring for compact embedded integration.

TECHNICAL SKILLS

Software & Tools: Python | C++ | ROS | MAVSDK | PX4 | Gazebo | OpenCV | MATLAB | JavaScript | SQL | MySQL | Snowflake | Power BI | Power Automate
AI & Productivity: Claude | ChatGPT | GitHub Copilot | Copilot Studio
CAD & Design: SolidWorks | Siemens NX | Revit | AutoCAD
Hardware & Lab: Embedded Systems | Soldering | 3D Printing | CNC Mill | Optical Comparator | Laser Marking (FOBA/MarkUS) | Data Acquisition | Prototyping
Certifications: CSWA (SolidWorks) | Siemens NX Certified

EDUCATION

Wentworth Institute of Technology Exp. Graduation: August 2027
B.S. Mechanical Engineering & Robotics Engineering Boston, MA
Minor: Aerospace Engineering | GPA: 3.75 | Presidential Scholarship | SAT: 1520 | ACT: 32
Coursework: Calculus I–III, Linear Algebra, Statistics & Probability, Thermodynamics, Fluid Mechanics, Dynamics, Mechanics of Materials, Embedded Systems, Robotics & Controls, Signals & Systems, Numerical Methods, Digital Logic, CAD & Solid Modeling